EXPERIMENT NO: 04

Experiment Title: Demonstrate JavaScript function types, scope, and closures; implement exception handling using try-catch; and develop a palindrome checker using functions.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Experiment Program Code

Code 1: Try_Catch.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h2>Try-Catch Example</h2>
    <p id = "result"></p>

    <script>
        try{
            document.getElementById("result").innerHTML = x;
        }
        catch(error){
            document.getElementById("result").innerHTML = "Error: "+error.message;
        }
    </script>
</body>
</html>
```

Code 2: Try_Catch_2.html.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <p id = "result"></p>

    <script>
        try{
            let num = 10;
            let result = num.toUpperCase();

            document.getElementById("result").innerHTML = result;
        }
        catch(error){
            document.getElementById("result").innerHTML = "An error occurred:" + error.message ;
        }
    </script>
</body>
</html>
```

Code 3: function_parameter.html

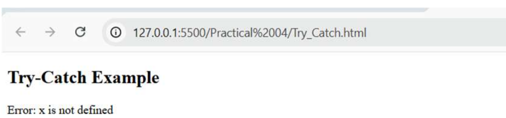
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Function Example </title>
</head>
<body>
    <h2>Addition Using Javascript function</h2>

    <input type='number' id='num1'><br><br>
    <input type='number' id='num2'><br><br>
    <button type='button' onclick='show()'>Add</button>

    <script>
        function add(a, b) {
            return a + b;
        }

        function show() {
            let result = add(
                Number(document.getElementById("num1").value),
                Number(document.getElementById("num2").value)
            );

            alert("The sum is : " + result);
        }
    </script>
</body>
</html>
```

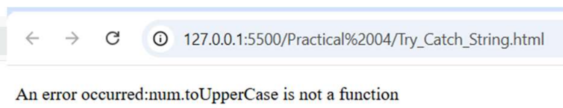
Code 4: Try_catch_prime_no.html

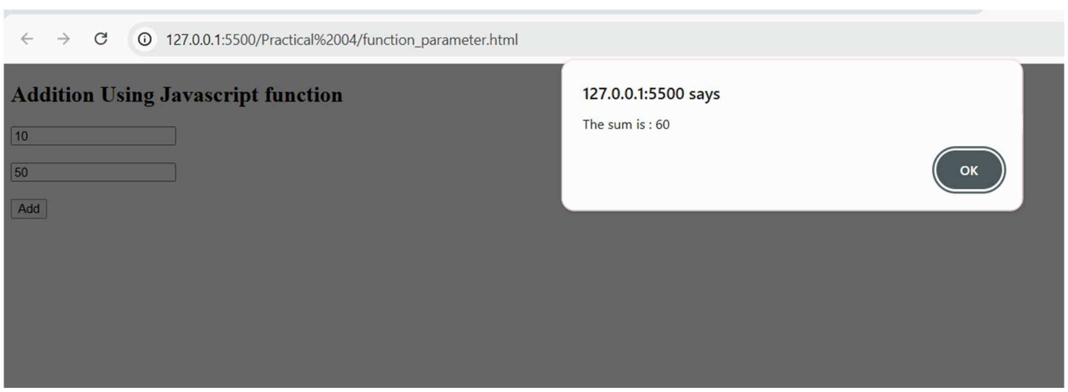
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Prime Number</title>
</head>

<body>

    <h2>Prime Number</h2>

    <p id="result"></p>

    <input type="text" id="box">
    <button onclick="checkPrime()">Check</button>

    <script>
        function checkPrime() {

            let num = document.getElementById("box").value;
            let result = document.getElementById("result");

            if (num === "") {
                throw new Error("Number cannot be empty.");
            }

            num = Number(num);

            if (isNaN(num)) {
                throw new Error("Please enter a valid number.");
            }

            if (num <= 1) {
                throw new Error("Number should be greater than 1.");
            }

            let prime = true;

            for (let i = 2; i < num; i++) {
                if (num % i === 0) {
                    prime = false;
```

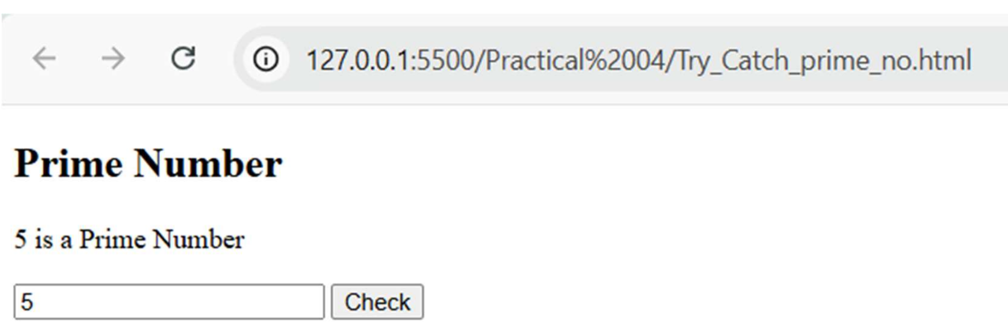
```
                    break;
                }
            }

            if (prime) {
                result.innerHTML = num + " is a Prime Number";
            } else {
                result.innerHTML = num + " is Not a Prime Number";
            }
        }
    </script>

</body>
</html>
```

4. Output









5. Case Study Title

Develop a palindrome checker using functions.

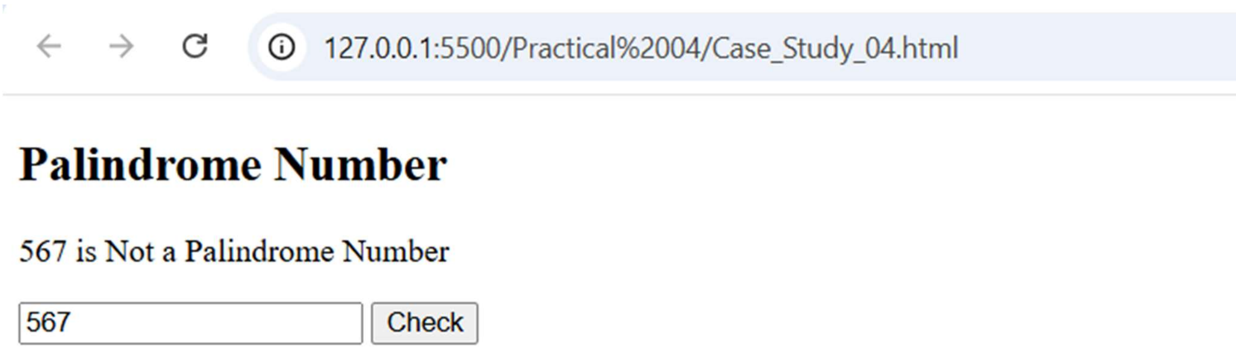
6. Case Study Program Code

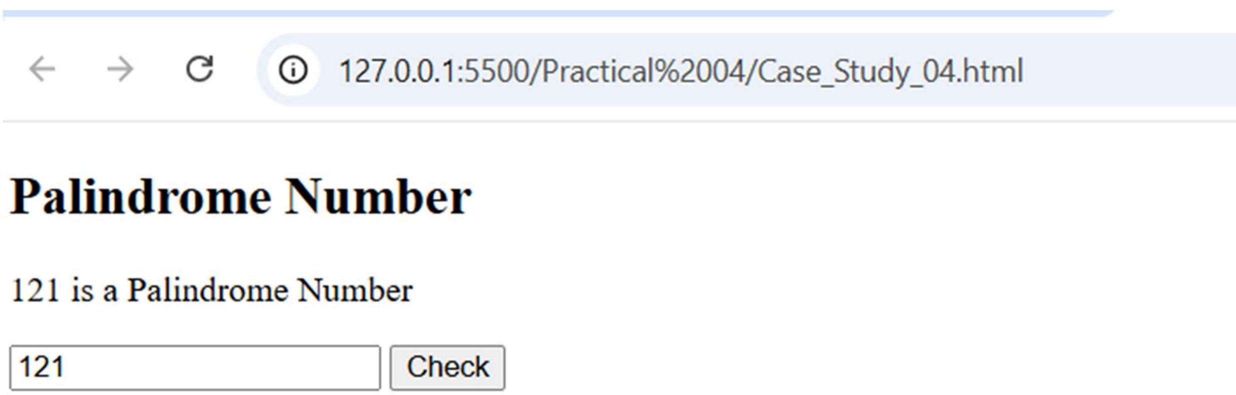
Casestudy_02_Kanak_Tembhurne_1290.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Palindrome Number</title>
</head>

<body>

    <h2>Palindrome Number</h2>

    <p id="result"></p>

    <input type="text" id="box">
    <button onclick="checkPalindrome()">Check</button>

    <script>

        function reverseNumber(num) {
            let reverse = 0;

            while (num > 0) {
                let digit = num % 10;
                reverse = reverse * 10 + digit;
                num = Math.floor(num / 10);
            }

            return reverse;
        }

        function checkPalindrome() {

            let num = document.getElementById("box").value;
            let result = document.getElementById("result");

            if (num === "") {
                throw new Error("Number cannot be empty.");
            }

            num = Number(num);

            if (isNaN(num)) {
                throw new Error("Please enter a valid number.");
            }

            if (num <= 0) {
                throw new Error("Number should be greater than 0.");
            }

            let reverse = reverseNumber(num);

            if (num === reverse) {
                result.innerHTML = num + " is a Palindrome Number";
            } else {
                result.innerHTML = num + " is Not a Palindrome Number";
            }
        }

    </script>

</body>
</html>
```

7. Output of the case study :

Name : Kanak Tembhurne PRN : 24070521290 File path :
http://127.0.0.1:5500/Practical%2002/Casestudy_04_Kanak_Tembhurne_1290.html





8. Result/Conclusion

Thus, the different types of JavaScript functions, variable scope, and closures were successfully demonstrated. Exception handling using try-catch and throw was also implemented to handle invalid inputs and runtime errors. A palindrome checker case study was successfully developed using a JavaScript function to reverse the given number and compare it with the original number, thereby determining whether the given number is a palindrome or not.